Effective and Robust: A Discriminative Temporal Learning Transformer for Satellite Videos

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***Abstract*— Robust feature learning has always been a research hotspot in dynamic temporal tasks. It makes the model almost unaffected by some challenging properties. The sequential nature of the transformer means attractive for temporal learning tasks, making it perform well in the video field. It is a current research hotspot for learning effective features by utilizing the target motion trends in satellite videos with multiple attributes, such as similar objects (SOBs) interference and occlusion. In this article, a novel discriminative temporal learning transformer tracker (DTLTracker) is introduced to characterize the dynamic target information for satellite videos. A discriminative transformer (DT) is proposed to comprehensively explore the dynamic target features with multiple attention mechanisms. It focuses on the primary information of the search area, making the target more discriminative. A fast convergence (FC) filter is designed to accelerate the weights convergence in calculating the target correlation operation, thereby ensuring the efficiency of model learning. The effectiveness and convergence have been demonstrated for the proposed optimization method. Additionally, a motion prior correction (MPC) module is constructed to utilize temporal information for target tracklet prediction, assisting the tracker in predicting the correct target. Numerous experiments are performed on three satellite videos to verify the effectiveness and feasibility of the proposed DTLTracker. It shows robustness compared to the state-of-the-art trackers on some challenging properties.**

***Index Terms*— Discriminative transformer (DT), filter learning, motion predictor, satellite video tracking.**

I. Introduction

EFFECTIVE feature learning enhances the adaptive representation capabilities of the model, which plays a crucial role in the dynamic satellite scenarios [1], [2], [3], [4], [5], [6]. The continuous observation of the satellite to the ground forms the high-resolution satellite video data [7], [8], [9], [10]. It has the characteristics of high resolution with frame image and small proportion with tracked target pixels [7], [11], [12]. The vehicles' targets with similar appearance make the tracker difficult to distinguish the target position across the video sequences. The low contrast between the foreground and background leads to the blurred target boundaries [13], [14], [15], [16]. The model may be affected by target shadows or noise. It requires to utilization of temporal context information effectively. In addition, the target occlusion caused by the external environment, such as bridge piers and trees, trials the accuracy of the model prediction on the dynamic target trajectory.

Satellite video tracking plays an important role in military activities and urban traffic detection [4], [17], [18], [19], [20]. Given a target bounding box of the first frame from a video, the main purpose of this task is to annotate the tracked box in the subsequent video sequence [21]. Many siamese-based satellite object trackers have appeared after Siamese-FC proposed [18], [19], [20], [22], [23]. They take two samples as inputs, which generally appear as the template region in the first frame and the search area in the current frame. The model then computes the object location for the current frame by comparing the similarity of the two inputs across high-dimensional space. Some trackers adopt the multihead cross-attention module for obtaining the response map in the cross correlation [24], [25]. However, they still need to be improved in the utilization of temporal context information during online processing. The correlation filters based on the Siamese network take advantage of the convolutional network to extract deep features [26], [27], [28]. They obtain the discriminative target localization with filters iterated by the steepest descent method in the online tracking phase.

Transformers have been shown suitable for video tasks because of the video sequential properties [29], [30], [31], [32]. They build long-term relationships effectively in the video sequence as encoder–decoder modules. Transformer trackers use the transformer architecture to aggregate the features across different branches in the Siamese network [29], [30]. They have achieved effective target feature enhancement. The visual tracker can be transferred to the remote sensing field, as shown in Fig. 1 of TrDiMP [29] whose sensitive to intraclass objects. It is necessary to consider the similar foreground and clutter background interference in dynamic satellite scenes. Our goal is to establish a model specific to satellite scenes that is discriminative for features.

By discovering the problems encountered by transformers in the satellite video tracking field, we find that adjacent intraclass objects are not well suppressed. On the other hand, dynamic videos contain rich target information, which

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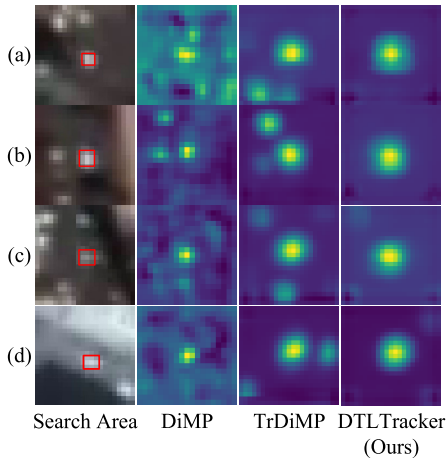


Fig. 1. Tracking confidence maps of DiMP [27] (second column), TrDiMP [29] (third column), and the proposed DTLTracker (last column). (a)–(d) Sequence 5, 31 and 33 in Video 1, and Sequence 17 in Video 2 of SV248S [7], respectively.

involves not only spatial information but also temporal motion information. It is crucial for target feature learning with accurate tracking. Some trackers [5], [33] have attempted to use Transformers to process dynamic spatiotemporal information. Others constructed an inference network based on an attention mechanism to suppress interference from similar targets [1], [18]. However, most of their focus is on dynamic feature learning or interference suppression without incorporating prior motion information of the target.

Therefore, a novel discriminative temporal learning transformer tracker (DTLTracker) is developed to learn the target temporal information robustly in satellite videos. Taking inspiration from video transformers, a discriminative Transformer (DT) is constructed with multiple discriminative attention. It not only suits satellite scenes with limited spatial information but also captures spatiotemporal contextual clues of the target. The convergence trend of the discriminative correlation filter in the baseline is slow in the dynamic target feature learning process. Therefore, a fast convergence (FC) filter optimization is proposed based on the conjugate gradient method [34]. We have proved the objective function with the proposed optimization achieves global convergence after the iteration ends. Temporal information not only includes dynamic feature change, the movement trend of the tracked target. A motion prior correction (MPC) module is proposed to guide the tracker for successfully tracking the target on challenging attributes. The confidence score calculated by the proposed FC filter is combined with the updated Kalman filter to form the correction value. Fig. 1 shows the confidence maps of our DTLTracker and other Siamese-based filter trackers. Similar objects (SOBs) or background interference have been suppressed effectively among the proposed DTLTracker. We evaluate the DTLTracker performance on several public satellite datasets, including SV248S [7], VISO [11], and SatSOT [12]. Experimental results show that the performance of our model can be comparable to the state-of-the-art trackers. The main contributions of this article can be summarized as follows.

1) A DTLTracker is developed to integrate the target spatiotemporal characteristics. It allows the model to be free from background and SOB interference.
2) We propose a DT for learning the dynamic features robustly. It utilizes a novel fusion decoder containing multiple discriminative attention mechanisms to enhance feature diversity, which endows the ability to learn the key features.
3) A FC filter based on the conjugate gradient method is adopted without additional parameters. It achieves rapid optimization of the parameters by calculating the step length and conjugate direction iteratively. A convergence proof of its optimization is given for the objective function.
4) A MPC model is proposed to correct the learning strategy during the online tracking phase timely. It combines the results of the fast filter and the Kalman filter, which performs stably on the attributes that SOB interference and target reappear after occlusion.

The remainder of this article is organized as follows. In Section II, the work related to our proposed method is presented. Section III provides the details of the proposed transformer tracker. Section IV verifies the feasibility and effectiveness of the model in satellite video tracking tasks. This article is concluded in Section V.

II. Related Works

A. Visual Object Tracking

The trackers are roughly divided into estimation-based correlation filter methods and neural network-based deep learning methods. The formers fuse the hand-crafted or/and convolutional features based on the correlation filters [35], [36], [37], [38], [39]. But they incorporate surrounding background noise during the circulant shift process, which makes the positive samples obtained during model training larger than the ground truth [35], [36], [37]. The later discriminative tracker has become a robust and efficient framework in video tracking by learning background and target features online in the Fourier domain [38], [40]. Among them, BACF adds background patches to the training of the objective function, effectively enhancing the target perception. STRCF introduces a temporal regularization term to avoid overfitting by emphasizing the recent samples. Some introduce spatial regularization terms to achieve discriminative processing of multichannel features. However, focusing solely on the effective learning of filters is far from enough, which limits the diversity of feature extraction.

Deep learning-based tracker relies on pretrained network trained from detection data as the backbone extractor, followed by a classifier to achieve target localization [41], [42], [43], [44], [45]. MDNet [41], as the classic model for the single-branch tracker, trains a fully connected layer specific to the current sequence across the first frame target information for subsequent target tracking. But it is not sensitive to intraclass objects, which needs improvement under SOB interference. On this basis, RT-MDNet [42] introduces an improved RoIAlign method to optimize target positioning. It learns an embedding space for distinguishing targets with similar semantic information while tracking failed missed appearance information, such as complete occlusion. To this

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end, we introduce an MPC mechanism with multiple indicators for online processing of tracking failures caused by SOB interference or occlusion. It improves both model performance and running speed.

Siamese-based trackers determine the target location by calculating the similarity between the search area and the template features [43], [44], [46], [47]. CGACD utilizes correlation-guided attention modules to enhance the features of corner regions, thereby improving detection accuracy. SiamGAT [48] focuses on developing an improved similarity calculation, it yields a response map through multihead cross attention instead of a simple cross correlation operation. However, they have limited features learned from low-resolution satellite targets, resulting in a lower performance. DiMP [27] develops a model optimizer to learn background and foreground information. It accelerates convergence through the steepest descent methodology under the least-squares loss. Its optimization-based Siamese network can be well migrated to the satellite video field. However, we find that the descent speed of this convergence method needs to be improved through experimental observation. Based on the above analysis, we introduce a novel FC optimization strategy to avoid calculating the Hessian matrix without any additional parameters.

B. Video Transformers

The video transformers learn dynamic scenes by building global context in temporal information modeling [31], [32], [49], [50], [51]. Attention, as an effective part of spatiotemporal learning, plays an important role in the temporal Transformer domain. SAAN [52] adopts an attention mechanism guided by feature similarity to enhance channel and spatial feature representation. It demonstrates the strong ability to capture the information similarity relationship. STM [50] utilizes intermediate clues of frames to cope with foreground appearance changes. Its form of attention-based cross correlation calculation between memory information and current frame information uses more temporal information. It provides an effective reference value for the correlation calculation paradigm between the current frame and the previous frames. AOT [53] draws on the memory reading paradigm of STM and designs long-term and short-term attentions to achieve hierarchical matching and propagation. It effectively enhances the target correlation between long-range and short-range. Inspired by the memory utilization in video tasks, we introduce a feature memory module after the proposed spatial transformer encoder to store the spatial–temporal information of the target.

Transformers have demonstrated strong abilities to utilize spatial-temporal information in visual object tracking. TransT [54] utilizes self-attention mechanism to enhance features and replaces the correlation calculation with cross-attention to establish more semantic feature maps than correlation. While TrTr [30] performs cross-attention operations on the search area features. It obtains global and rich contextual interdependencies by applying a transformer encoder–decoder. TrDiMP [29] designs two cross-attention mechanisms for enhanced search features, which aggregate more valuable target temporal information while aiding in target localization. Taking inspiration from them, we designed a DT tracker constructed with multiple cross-attentions. From the last two columns of Fig. 1, it can be observed that the proposed tracker can remedy the contextual noise while enhancing the learning of the target itself. However, experiments have found that these trackers fail to fully utilize the moving target tracklet for prediction when target features cannot be learned, such as reappearance after occlusion. We adopt an MPC scheme based on a target tracklet, which effectively avoids tracking failure when the target reappears after occlusion.

C. Object Tracking for Satellite Videos

The current trackers transformed the visual object tracking methods to the satellite video fields. They have improved at the original trackers for adapting to the remote sensing scenes [4], [20], [23]. The correlation filter trackers extract positive and negative samples from the given target box in the first frame for training. They update the weights regularly, as well as achieve real-time and adaptive tracking without pretraining, which is suitable for satellite scenarios. In addition, their movements are usually regular, such as predicting the current target position by analyzing previous motion trajectories. Some trackers use motion estimation (ME) to assist the correlation filter for adapting the remote sensing scenarios [4], [55], [56], [57]. Among them, the Kalman filter has high accuracy in evaluating object motion, which is often used to estimate the position and velocity of the tracked target [56], [57]. CFME [56] adopts the ME method composed of the Kalman filter and motion trajectory averaging method to integrate the target motion characteristics. In the occlusion problem, the response patch peak of the filter is used to determine whether the target is occluded and the occlusion ends. If occluded, the ME result is used as the target position. Taking inspiration from this, we use the Kalman filter to achieve target tracklet prediction. However, the dynamic characteristics of targets have not been fully utilized based on correlation filters. We utilize an attention-based transformer encoder–decoder to make the tracker focus on the foreground target rather than background noise in this article. The fusion of multiple attentions achieves feature diversity.

Siamese networks achieve end-to-end learning by matching similarity between templates and search regions based on pretrained backbones [5], [19], [20], [22], [58]. They learn more discriminative features compared to the correlation filter trackers. It is particularly important to make full use of spatiotemporal information in satellite videos with a lack of appearance information and dense similar foregrounds. HRSiam [20] uses HRNet to extract high-resolution features from input patches. It develops an adaptive fusion strategy by fusing the Gaussian mixture model to detect moving targets in pixels. Its learning of the target temporal information is insufficient, resulting in less robust performance when dense similarity situations occur. Li et al. [5] propose a transformer tracker with a complex-former position encoding method to mine the spatial dependence of distance and direction. It increases computational complexity to a certain extent with

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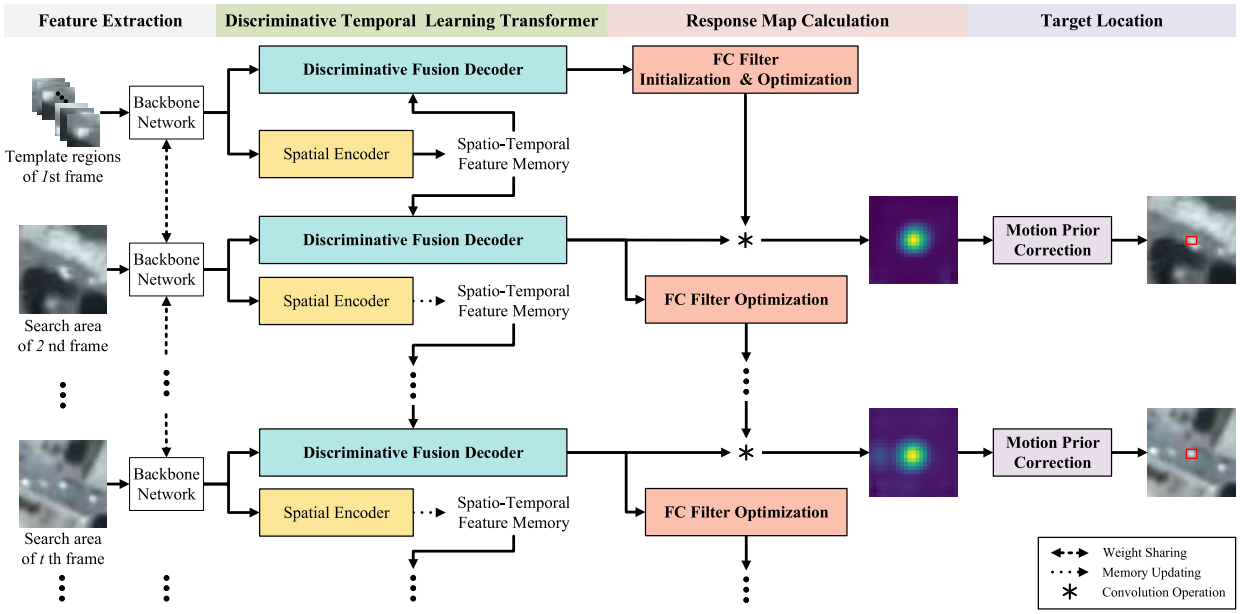


Fig. 2. Illustration of the proposed DTLTracker for online tracking. The discriminative semantic features output by the proposed decoder are convolved with the FC filter to obtain a confidence map. The MPC module in the post-processing stage is employed for the tracklet correction.

a greater room for development in learning target temporal features. For this purpose, we design a novel Transformer based on multiple cross-attention mechanisms to learn target spatiotemporal features in a video sequence. It captures the different correlations of the tracked target information, and provides an effective memory function for the untrained tracked target in the online tracking stage.

III. Proposed Approach

In this section, we elaborate on the structure of DTLTracker in dynamic sequence processing. The overall architecture of online learning is first introduced to have a clear understanding of the proposed framework. The proposed DT is then presented, which contains a spatial encoder and a discriminative fusion decoder. The theoretical derivation of the fast filter optimization is introduced in Section III-C. Finally, the MPC module is employed to regulate the tracking trajectory.

A. Tracking Framework

The temporal relationship across frames is the key instance for video learning. We adopt the proposed DTLTracker at satellite videos to learn the target information from specific sequences. It enables the tracker to reduce interference from the background and similar intraclass objects. As displayed in Fig. 2, the mainstream tracking model is formulated into the Siamese network. We employ a hybrid model with CNN and transformer. The tracker adopts the convolutional blocks in the shallow layer and the proposed transformer in the deep layer to enhance the modeling ability.

The top branch learns the model initialized by the templates, the bottom branches focus on filter optimization and target localization. Multiple encoded template features in the first frame are sent to the discriminative fusion decoder successively for implementing the feature learning and the filter optimization operations. In the normal status, the encoded region features are fed into the memory $\tilde{X}$. The update scheme in Fig. 2 shows that the target features stored in the memory affect the filter optimization and target location. $\tilde{X}$ will be updated under normal status to learn scenario changes adaptively. A queue storage mode is developed to control the memory within a certain range. In this way, storage can be avoided excessively with low computation costs.

The filter parameters are designed with the target information in a specific sequence. Its optimization is adaptively based on the fused decoded features. The initial template features lay the foundation for its optimization. In addition, the MPC module, as an effective embedding way, can simulate human perception of object motion in the real world. It is suitable for dealing with situations such as object occlusion and SOB interference.

B. Discriminative Transformer

Transformers have been widely explored in the dynamic learning of object features in videos [47], [53], [59], [60], [61], [62], [63]. We take the template/search features from the backbone as inputs to the proposed transformer. The feature field is set in a predefined fixed pattern to simplify the attention matrix. The proposed transformer architecture is shown in Fig. 3, where the spatial encoder consists of a self-attention mechanism and a layer-normalization with the residual connection module. The semantic features from dynamic temporal sequences are stored in the spatiotemporal feature memory module after passing through the encoder. Particularly, a discriminative fusion decoder is designed to adaptively learn the dynamic target characteristics in the online learning phase. It enables the model to be discriminatively free from SOB interference.

1) Spatial Encoder: This module learns the template feature, whose output is the object feature storage of the memory.

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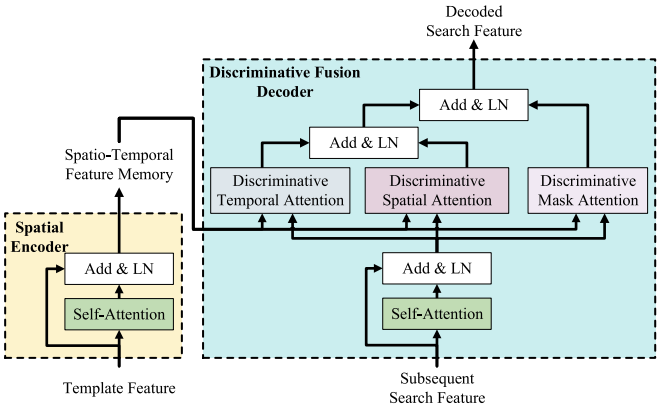


Fig. 3. Proposed DT architecture.

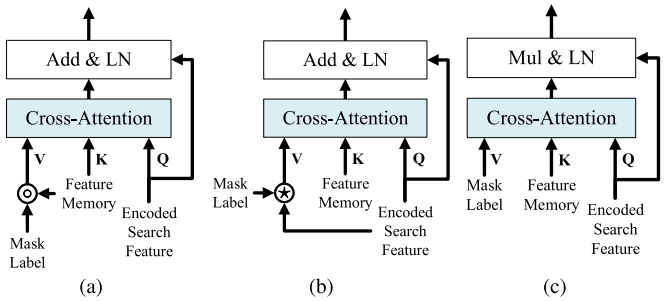


Fig. 4. Different attention modules. (a)–(c) Discriminative temporal, spatial, and mask attention modules, respectively.

It establishes the effectiveness of the subsequent object representation learning. The encoder block contains a self-attention mechanism and a residual normalization module, which can be described as

$$\hat{X} = \mathrm{LN}(A_X + X). \tag{1}$$

Given a set of N template regions, the backbone network generates the transformed convolution features $\breve{X} \in \mathbb{R}^{N\times(S/H)\times H\times C}$ with spatial size $(S/H) \times H$ and channel number C. Here the input matrix X is the features $\breve{X}$ with flattened. The function $\mathrm{LN}(\cdot)$ represents the layer normalization. Its composition with a residual connection can smooth the loss and speed up the model convergence. The encoder output $\hat{X} \in \mathbb{R}^{NS\times C}$ is stored as the initial feature memory. Inspired by the space-time memory networks in video object segmentation [50], we use the spatiotemporal feature memory $\tilde{X} \in \mathbb{R}^{NS\times C}$ to store the dynamic features. As the dynamic storage of the encoder output and the decoder input, the feature memory plays an important role in temporal feature learning. It provides a guarantee for the attention mechanism calculation among the proposed decoder and enables the model to discriminate for the target features.

The self-attention mechanism output $A_X \in \mathbb{R}^{NS\times C}$ determines the weight of each input across the mutual game between the inputs. The attention function focuses on the relevant part of each input feature. It captures the correlations within the template features, which can be thought of as dense relationships within the features. The mechanism is calculated as

$$A_X = \mathrm{Softmax}\left(Q_X K_X^T\right) V_X \tag{2}$$

where the $\mathrm{Softmax}(\cdot)$ is used as a weight assignment function to measure the similarity between Q_X and K_X as well as prevent the gradient from disappearing. Besides, the numerical distribution of the matrix $Q_X K_X^T$ processed by this function is relatively scattered, which is suitable for the requirements of model training. The multiplication of two matrices indicates the importance across different positions. The larger the element value, the higher the similarity. Here $Q_X = XW_{\mathrm{En}}^q$, $K_X = XW_{\mathrm{En}}^k$, and $V_X = XW_{\mathrm{En}}^v$ are intermediate representations of the input feature X. That is, the attention mechanism inputs are obtained by mapping the feature to different spaces. It enhances the model expression ability as well as improves generalization ability. $W_{\mathrm{En}}^q \in \mathbb{R}^{C\times w^q}$, $W_{\mathrm{En}}^k \in \mathbb{R}^{C\times w^k}$, $W_{\mathrm{En}}^v \in \mathbb{R}^{C\times w^v}$ are the weight matrices which commonly stand for the query, key and value projections, respectively.

2) Discriminative Fusion Decoder: Temporal information is an important aspect of the video processing field. We design a discriminative fusion decoder on the right side of Fig. 3 to facilitate object feature learning for dynamic scenes effectively. It contains self-attention, cross-attention, and residual normalization modules. This decoder strengthens the spatiotemporal information transfer through the fusion of different cross-attention modules. It completes the aggregation of feature information diversity, which can be expressed as

$$\bar{Y} = \mathrm{LN}(\mathrm{LN}(\bar{Y}^{\mathrm{DTA}} + \bar{Y}^{\mathrm{DSA}}) + \bar{Y}^{\mathrm{DMA}}) \tag{3}$$

here $\bar{Y} \in \mathbb{R}^{S\times C}$ is the final decoded feature. $\bar{Y}^{\mathrm{DTA}}$, $\bar{Y}^{\mathrm{DSA}}$, and $\bar{Y}^{\mathrm{DMA}}$ represent different calculation types between the feature memory $\tilde{X}$, the encoded search feature $\hat{Y} \in \mathbb{R}^{S\times C}$, and the mask labels $M \in \mathbb{R}^{N\times(S/H)\times H}$. They are the feature outputs of the discriminative temporal, spatial, and mask attention modules, respectively. The decoder synthesizes the spatial–temporal context information with object features in dynamic temporal sequence through cascade combination. The proposed transformer achieves effective temporal feature learning with enhancing characterization capabilities through fusing multiple attention forms.

a) Discriminative temporal attention: The cross-attention mechanism plays an important role in learning and aggregating features. As shown in Fig. 4(a), it computes the cross-attention between search features and memory, which is as follows:

$$\bar{Y}^{\mathrm{DTA}} = \mathrm{LN}(A^{\mathrm{DTA}} + \hat{Y}) \tag{4}$$

where $\hat{Y} = \mathrm{LN}(A_Y + Y)$ is the encoded search feature via a self-attention module, such as (1). Here $Y \in \mathbb{R}^{S\times C}$ is the flattened search convolutional feature, and A_Y is the output feature encoded by the self-attention module. The cross-attention value can be formulated as follows:

$$A^{\mathrm{DTA}} = \mathrm{Softmax}(Q_{\hat{Y}} K_{\tilde{X}}^T) V_{M_C \circ \tilde{X}}. \tag{5}$$

This attention matrix $\mathrm{Softmax}(\cdot)$ is based on the calculation of memory and search features. It captures the discriminative target temporal context dependence with the reduced calculation

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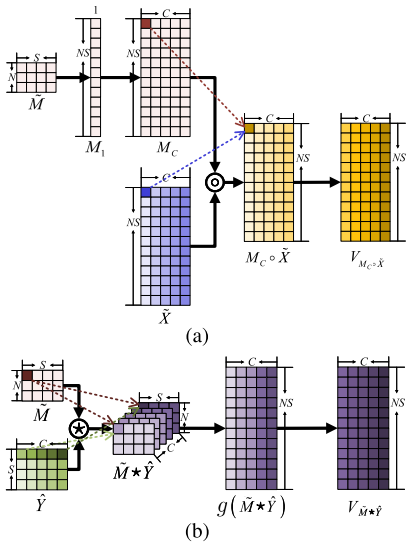


Fig. 5. Detailed diagrams of value variants. N, S, NS, and C represent the dimensionality of the corresponding feature matrix. Below the feature matrix are the corresponding variables mentioned in this article. (a) and (b) Value input of the discriminative temporal and spatial attention modules, respectively.

amount by providing a local receptive field. The query, key, and value elements are represented in the following forms:

$$\begin{cases} Q_{\hat{Y}} = \hat{Y} W_1^q, \\ K_{\tilde{X}} = \tilde{X} W_1^k, \\ V_{M_C \circ \tilde{X}} = (M_C \circ \tilde{X}) W_1^v \end{cases} \tag{6}$$

where $\circ$ means the Hadamard product. $Q_{\hat{Y}} \in \mathbb{R}^{S \times w_1^q}$, $K_{\tilde{X}} \in \mathbb{R}^{NS \times w_1^k}$, and $V_{M_C \circ \tilde{X}} \in \mathbb{R}^{NS \times w_1^v}$ are defined as the query, key, and value, respectively. $W_1^q \in \mathbb{R}^{C \times w_1^q}$, $W_1^k \in \mathbb{R}^{C \times w_1^k}$, and $W_1^v \in \mathbb{R}^{C \times w_1^v}$ are the corresponding weighted matrices. In this article, $w_1^q = w_1^k < C$ and $w_1^v = C$ are employed as default values.

We have marked the size of each matrix in Fig. 5 for easy observation. The value $V_{M_C \circ \tilde{X}}$ in Fig. 5(a) realizes the effective utilization of the template information. Here $\tilde{M} \in \mathbb{R}^{N \times S}$ represents the mask matrix M flattened in space resolution. $M_1 \in \mathbb{R}^{NS \times 1}$ represents the flattened $\tilde{M}$ with channel number of 1. The mask matrix $M_C \in \mathbb{R}^{NS \times C}$ is devised the same channel number as the memory $\tilde{X}$ to implement the element-wise product. The original mask matrix M is designed as the Gaussian map of the previous box. It makes the model learn the target-specific features and focus more on the target area. Besides, we set the mask and memory quantity to the same number N for product calculation.

b) Discriminative spatial attention: The aggregation between current features and target information can enhance the model's ability to perceive dynamic features. A discriminative spatial attention module is designed to capture the spatial semantic information. The overall attention calculation formula $\bar{Y}^{\mathrm{DSA}} = \mathrm{LN}(A^{\mathrm{DSA}} + \hat{Y})$ combines the current features with a greater extent. Here $A^{\mathrm{DSA}} = \mathrm{Softmax}(Q_{\hat{Y}} K_{\tilde{X}}^T) V_{\tilde{M} \star \hat{Y}} \in \mathbb{R}^{S \times C}$ is similar to (5). It enhances the feature expression ability at the pixel level. The difference is the value matrix, whose calculation formula is

$$V_{\tilde{M} \star \hat{Y}} = g(\tilde{M} \star \hat{Y}) W_2^v \tag{7}$$

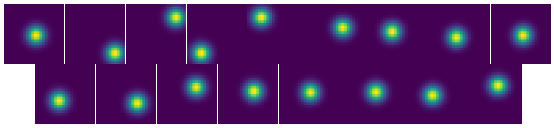


Fig. 6. Transformed mask labels.

here $W_2^v \in \mathbb{R}^{C \times w_2^v}$ is trainable parameter of the projection. $g : \mathbb{R}^{N \times S \times C} \to \mathbb{R}^{NS \times C}$ is a reshape operation. The matrix $\tilde{M} \star \hat{Y} \in \mathbb{R}^{N \times S \times C}$ is designed as the product of the mask maps and the encoded search feature. It can be calculated by

$$(\tilde{M} \star \hat{Y})_{n,s,c} = \tilde{m}_{n,s} \hat{y}_{s,c} \tag{8}$$

where $\tilde{M} = (\tilde{m}_{n,s}) \in \mathbb{R}^{N \times S}$. $\hat{Y} = (\hat{y}_{s,c})$ is the encoded search feature. $n \in [1, N]$, $s \in [1, S]$ and $c \in [1, C]$ are expressed as the element subscripts of the corresponding matrixes. The specific calculation of value is shown in Fig. 5(b), which enables a feature matching between search area information and mask maps. This value performs dot product calculations on the current feature and the object mask information across different channels. It achieves a weighted fusion among spatial and channel information. Different from the temporal memory characteristics on $V_{M_C \circ \tilde{X}}$ in Section III-B2.a, this attention module is adaptive to the current spatial information. It makes the model robust to the target temporal feature learning.

c) Discriminative mask attention: We aim to improve the discriminative learning of the tracked target features. Different from the previous two attention modules, this mask attention employs the Hadamard product instead of the adding operation in residual normalization. It generates a mask attention using mask values as input. The attention calculation is specifically expressed as follows:

$$\bar{Y}^{\mathrm{DMA}} = \mathrm{LN}(A^{\mathrm{DMA}} \circ \hat{Y}) \tag{9}$$

where $A^{\mathrm{DMA}} = \mathrm{Softmax}(Q_{\hat{Y}} K_{\tilde{X}}^T) V_{M_C} \in \mathbb{R}^{S \times C}$ is the attention matrix. $V_{M_C} = M_C W_3^v \in \mathbb{R}^{NS \times C}$ denotes the value input. Here M_C is the same as the mask input before the product at Fig. 5(a). $W_3^v \in \mathbb{R}^{C \times w_3^v}$ represents the weight matrix with $w_3^v = C$. As for the masking label M, some 2-D Gaussian distributions are constructed with different augmentation values, where the target center is the mean value. It can clearly indicate the key areas where the model focuses. The mask element m_n is expressed as

$$m_n(z) = e^{-\frac{(z-b_n)^2}{2\sigma^2}} \tag{10}$$

here z is the position in the mask matrix. σ is the variance, whose related to the spatial size S. b_n represents the nth augmentation value based on the target center position. The mask label augmentation operations are supervised geometric transformations, including flip, rotation, blur, dropout, etc. The diversity mask labels are shown in Fig. 6. These transformed mask labels help to improve the robustness and feature diversity of modeling.

This module can be regarded as a mask operation across the search features, which has lower computational complexity than the previous two modules. It only supervises local information to make a simple prediction, ensuring the mask is

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not involved in attention processing. Besides, this attention can effectively avoid the impact of clutter information, so that the model generates information whose spotlighted on the object features.

C. FC Filter

We design a filter f to implement target appearance learning. This filter model consists a set of training sample set $\mathfrak{S} = \{(\bar{Y}_i, \mathfrak{C}_i)\}_{i=1}^{\mathcal{N}}$ after extracting by the discriminative decoder. $\mathfrak{C} = (l, r)$ represents the target center coordinates corresponding to the decoder feature $\bar{Y}$. The loss function of the filter f is written as follows:

$$\mathcal{L}(f) = \frac{1}{\mathcal{N}} \sum_{(\bar{Y},\mathfrak{C})\in\mathfrak{S}} \|\mathfrak{r}(\bar{Y} * f, \mathfrak{C})\|^2 + \|\lambda f\|^2 \tag{11}$$

here $\mathfrak{r}(\cdot)$ represents the residual function of the two input coordinates inspired by the least-squares-based regression. λ is the regularization term. The loss function $\mathcal{L}(\cdot)$ is fit to the second-order Taylor expansion at the ith iteration weights f_i as gives

$$\begin{aligned}\mathcal{L}(f) \approx \bar{\mathcal{L}}(f_i) &= \mathcal{L}(f_i) + (f - f_i)^T \nabla\mathcal{L}(f_i) \\ &\quad + \frac{1}{2}(f - f_i)^T \frac{\partial^2\mathcal{L}}{\partial f^2}\bigg|_{f_i} (f - f_i)\end{aligned} \tag{12}$$

where $\bar{\mathcal{L}}(f_i)$ represents the Taylor expansion function at f_i. $\nabla\mathcal{L}(f_i) = (2/\mathcal{N})\sum((\partial\mathfrak{r}/\partial f)|_{f_i})^T\mathfrak{r} + 2\lambda^2 f$ is the gradient, and $(\partial^2\mathcal{L}/\partial f^2)|_{f_i} = (2/\mathcal{N})\sum((\partial\mathfrak{r}/\partial f)|_{f_i})^T(\partial\mathfrak{r}/\partial f)|_{f_i} + 2\lambda^2 I$ is the second derivative.

For converging quickly to the optimal parameters during the tracking phase, the baseline [27] uses the steepest descent methodology with the Gauss–Newton method for achieving the weights to converge. However, experiments have found that this convergence speed still needs to be improved. We adopt a conjugate gradient-based filter optimization in this article. It optimizes the filter with no additional parameters, avoiding the calculation of the Hessian matrix $(\partial^2\mathcal{L}/\partial f^2)|_{f_i}$.

The iterative strategy for the filter is $f_{\mathrm{i}+1} = f_i + \alpha_i d_i$, where α_i is the step length. The optimization direction $d_i = -\nabla\mathcal{L}(f_{\mathrm{i}}) + \beta_{\mathrm{i}-1}d_{\mathrm{i}-1}$ is conjugate with the previous direction. It is a linear combination of the gradient and the search direction, which does not affect the weights of the previous parameters $f_{\mathrm{i}-1}$. Here we set $\beta_i = (\nabla\mathcal{L}(f_{\mathrm{i}+1})^T\nabla\mathcal{L}(f_{\mathrm{i}+1}))/(\nabla\mathcal{L}(f_{\mathrm{i}})^T\nabla\mathcal{L}(f_{\mathrm{i}}))$ to be the Fletcher–Reeves formula. The first search direction d_0 is defined as the opposite direction of the gradient $\nabla\mathcal{L}(f_0)$. The process ensures the filter is optimized to the minimum value in each direction. Besides, the gradient can be expressed as $\nabla\mathcal{L}(f_i) = (\partial^2\mathcal{L}/\partial f^2)|_{f_i} f_i + \varepsilon$. We get $\nabla\mathcal{L}(f_{\mathrm{i}+1}) - \nabla\mathcal{L}(f_i) = (\partial^2\mathcal{L}/\partial f^2)|_{f_i}(f_{\mathrm{i}+1} - f_i) = \alpha_i(\partial^2\mathcal{L}/\partial f^2)|_{f_i} d_{\mathrm{i}}$. α_i then can be given as follows by multiplying the above formula by d_{i}^T:

$$\begin{aligned}\alpha_i &= \frac{d_{\mathrm{i}}^T(\nabla\mathcal{L}(f_{\mathrm{i}+1}) - \nabla\mathcal{L}(f_i))}{d_{\mathrm{i}}^T \frac{\partial^2\mathcal{L}}{\partial f^2}\big|_{f_i} d_{\mathrm{i}}} \\ &= \frac{-d_{\mathrm{i}}^T\nabla\mathcal{L}(f_i)}{d_{\mathrm{i}}^T \frac{\partial^2\mathcal{L}}{\partial f^2}\big|_{f_i} d_{\mathrm{i}}} = \frac{\nabla\mathcal{L}(f_i)^T\nabla\mathcal{L}(f_i)}{d_{\mathrm{i}}^T \frac{\partial^2\mathcal{L}}{\partial f^2}\big|_{f_i} d_{\mathrm{i}}} \\ &= \frac{\|\nabla\mathcal{L}(f_i)\|^2}{\frac{2}{\mathcal{N}}\sum\left(\frac{\partial\mathfrak{r}}{\partial f}\big|_{f_i} d_{\mathrm{i}}\right)^T\left(\frac{\partial\mathfrak{r}}{\partial f}\big|_{f_i} d_{\mathrm{i}}\right) + 2\lambda^2 d_{\mathrm{i}}^T d_{\mathrm{i}}} \\ &= \frac{\|\nabla\mathcal{L}(f_i)\|^2}{\frac{2}{\mathcal{N}}\sum\|\frac{\partial\mathfrak{r}}{\partial f}\big|_{f_i} d_{\mathrm{i}}\|^2 + 2\lambda^2\|d_{\mathrm{i}}\|^2}.\end{aligned} \tag{13}$$

Lemma 1: Given the objective function $\mathcal{L}(f)$, the Fletcher–Reeves conjugate gradient optimization [34], [64] with a restart strategy ensures that the function reaches global convergence after a certain iteration.

Proof: We assume that the exact solution $\mathcal{L}(f_{\mathrm{i}-1})$ has been obtained at the i−1th iteration. That is, $\alpha_{\mathrm{i}-1}$ is the optimal step length of the conjugate direction $d_{\mathrm{i}-1}$. The gradient $\nabla\mathcal{L}(f_{\mathrm{i}})$ is perpendicular to the i − 1th iteration direction $d_{\mathrm{i}-1}$. Thus, we have $\nabla\mathcal{L}(f_i)^T d_{\mathrm{i}-1} = 0$

$$\begin{aligned}\nabla\mathcal{L}(f_i)^T d_{\mathrm{i}} &= \nabla\mathcal{L}(f_i)^T(-\nabla\mathcal{L}(f_{\mathrm{i}}) + \beta_{\mathrm{i}-1}d_{\mathrm{i}-1}) \\ &= -\|\nabla\mathcal{L}(f_i)\|^2 + \beta_{\mathrm{i}-1}\nabla\mathcal{L}(f_i)^T d_{\mathrm{i}-1} \\ &= -\|\nabla\mathcal{L}(f_i)\|^2 < 0.\end{aligned} \tag{14}$$

The direction d_{i} in which the objective function decreases can be ensured to generate at each iteration i. The definition of the function $\mathfrak{r}(\cdot)$ is followed in the baseline [27]. It is linearly related to filter weights f. We have

$$\begin{aligned}\|\nabla\mathcal{L}(\dot{f}) - \nabla\mathcal{L}(\ddot{f})\| = \|\frac{2}{\mathcal{N}}\sum\frac{\partial\mathfrak{r}}{\partial f}^T\mathfrak{r}(\bar{Y} * (\dot{f} - \ddot{f}), \mathfrak{C}) \\ + 2\lambda^2(\dot{f} - \ddot{f})\| \le \delta\|\dot{f} - \ddot{f}\| \\ \forall \dot{f}, \ddot{f} \in f.\end{aligned} \tag{15}$$

The gradient $\nabla\mathcal{L}(f)$ has Lipschitz continuity property.

The restart strategy adopts the direction $-\nabla\mathcal{L}(f_0)$ of the steepest descent in the initial iteration stage. It ensures that the optimization restarts from the steepest descent direction. Here $(\partial^2\mathcal{L}/\partial f^2)|_{f_i}$ is a positive definite square matrix [27], which implements an exact line search at every restart. $\mathcal{L}(\cdot)$ is a second-order continuously differentiable function with a lower bound $\delta > 0$. It satisfies all prerequisites for the global convergence of conjugate gradient, so its last iteration point must be the stationary point of the objective function. □

In this filter iterative calculation, we do not need to calculate the second derivative, which is obtained through the neural network. Besides, the entire procedure of the filter optimization is given in Algorithm 1. The weights reach the global minimum point after the conjugate directions d_{i} is completed. Compared with the optimization based on the steepest descent methodology, this optimization-based structure can converge the weights to the extreme points quickly at the online learning stage. We perform visual analysis in the filter optimization part of Section IV-D.

D. MPC Scheme

The movement patterns of satellite targets can be set to static, uniform motion, or acceleration/deceleration motion. There is no overlap between them in satellite scenes. Under this premise, the second maximum score$_t^{\varphi_2}$ in the confidence map can be obtained after masking around the maximum score$_t^{\varphi_1}$, which is located at a certain distance from score$_t^{\varphi_1}$. They are defined as the tracked target or SOB. How to

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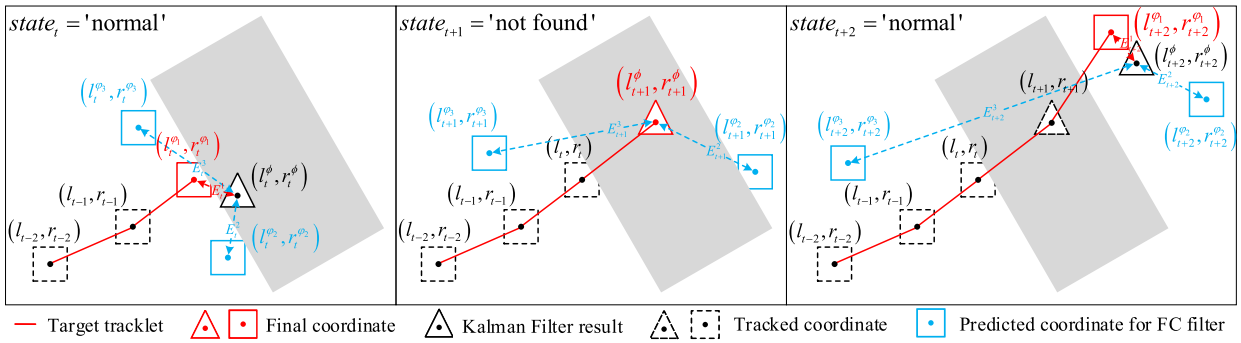


Fig. 7. Target tracklet schematic in consecutive frames. The blue and red dashed lines represent the Euclidean distance of coordinates predicted by the proposed FC and the Kalman filter.

Algorithm 1 Filter Optimization Process

Input: Initial filter parameters f_0, the training set $\mathfrak{S} = \{(\bar{Y}_i, \mathfrak{C}_i)\}_{i=1}^{\mathcal{N}}$

Output: Iterated parameters f

repeat
  if $\mathfrak{i} = 0$ **then**
    $\nabla\mathcal{L}(f_0) = \frac{2}{\mathcal{N}} \sum (\frac{\partial \mathfrak{r}}{\partial f}|_{f_0})^T \mathfrak{r} + 2\lambda^2 f_0$
    $d_0 = -\nabla\mathcal{L}(f_0)$
  else
    $\nabla\mathcal{L}(f_\mathfrak{i}) = \frac{2}{\mathcal{N}} \sum (\frac{\partial \mathfrak{r}}{\partial f}|_{f_{\mathfrak{i}-1}})^T \mathfrak{r} + 2\lambda^2 f_{\mathfrak{i}-1}$
    $\beta_{\mathfrak{i}-1} = {\|\nabla\mathcal{L}(f_\mathfrak{i})\|^2}/{\|\nabla\mathcal{L}(f_{\mathfrak{i}-1})\|^2}$
    $d_\mathfrak{i} = -\nabla\mathcal{L}(f_\mathfrak{i}) + \beta_{\mathfrak{i}-1} d_{\mathfrak{i}-1}$
  end if
  $\alpha_i = {\|\nabla\mathcal{L}(f_i)\|^2}/{(\frac{2}{\mathcal{N}} \sum \|\frac{\partial \mathfrak{r}}{\partial f}|_{f_i} d_\mathfrak{i}\|^2 + 2\lambda^2 \|d_\mathfrak{i}\|^2)}$
  $f_{\mathfrak{i}+1} = f_\mathfrak{i} + \alpha_i d_\mathfrak{i}$
until end of iteration

persuade the tracker to return to the right target under SOB interference is a challenge. In addition, stationary objects such as bridges, buildings, or trees often obstruct the target feature extraction. Currently, satellite trackers rely on embedding prior knowledge information, such as using road or motion priors [4], [56]. However, how to produce the target reappearing after occlusion is a challenge for the robust feature learning of the tracker. To this end, an MPC mechanism is proposed by fusing the results of the Kalman and FC filters. One indicator is the target state state_t obtained by the FC filter f. The other is the Euclidean distance between the results of the Kalman and FC filters.

For instance, in the current frame $t \in [1, T]$ at Fig. 7, We set the parameters of the Kalman and FC filter f to iterate once per frame. The final coordinate in the current frame after iteration is denoted as (l_t, r_t). The predicted target positions $\{(l_t^{\varphi_i}, r_t^{\varphi_i})\}_{i=1}^3$ calculated by the FC filter f and the decoded feature $\bar{Y}$ are as follows:

$$\begin{cases} \left[\text{score}_t^{\varphi_1}, \left(l_t^{\varphi_1}, r_t^{\varphi_1}\right)\right] = \max(\bar{Y} * f), \\ \left[\text{score}_t^{\varphi_2}, \left(l_t^{\varphi_2}, r_t^{\varphi_2}\right)\right] = \max\left(\text{mask}_{\varphi_1}(\bar{Y} * f)\right), \\ \left[\text{score}_t^{\varphi_3}, \left(l_t^{\varphi_3}, r_t^{\varphi_3}\right)\right] = \max\left(\text{mask}_{\varphi_2}\left(\text{mask}_{\varphi_1}(\bar{Y} * f)\right)\right) \end{cases} \tag{16}$$

here $\max(\cdot)$ function calculates the maximum score in the confidence map and its corresponding coordinate position. $\{\text{mask}_{\varphi_i}(\cdot)\}_{i=1}^2$ sets the map within a certain area around the position $(l_t^{\varphi_i}, r_t^{\varphi_i})$ to 0 for preventing the predicted target $(l_t^{\varphi_i}, r_t^{\varphi_i})$ from participating in the calculation.

On the other hand, the Kalman filter updates the parameters iteratively through the predicted measurement and estimated value. We use the initial target center coordinates as the initial filter state value. The state transition matrix is set to uniform, acceleration/deceleration motion, or stationary. The Kalman filter parameters are continuously iterated based on the target trajectory. The tracking results can be adjusted in a timely manner. Based on the confidence scores $\{\text{score}_{\varphi_i}\}_{i=1}^3$, correction values $\{E_t^i\}_{i=1}^3$ are designed to adjust the tracker learning strategy in a timely manner. It corrects the current predicted results $\{(l_t^{\varphi_i}, r_t^{\varphi_i})\}_{i=1}^3$ under SOB interference and the target reappearance after occlusion, represented by

$$E_t^i = \text{Euc}\left[\left(l_t^\phi, r_t^\phi\right), \left(l_t^{\varphi_i}, r_t^{\varphi_i}\right)\right] \tag{17}$$

here (l_t^ϕ, r_t^ϕ) is the coordinate predicted by the Kalman filter. Euc$[\cdot]$ represents the Euclidean distance between the coordinate results of the Kalman and FC filters. It avoids significant deviations in the target position during consecutive frames.

The model learns fewer effective target features in the search area after occlusion occurs, resulting in a lower confidence score. In this situation, the updated Kalman filter is activated by calculating the correction value E_t^i to predict the target motion trajectory. Stop updating the FC filter f simultaneously to avoid mistakenly adding background or noise to the parameters iteration. Otherwise, the wrong divided samples will affect the FC filter calculation for target features. At this point, the positive samples saved by the FC filter f are the tracked target information rather than cluttered backgrounds or noises. The premise for the target reappearance is defined as a lower correction value with a higher confidence score. The final target center coordinate (l_t, r_t) at the current frame is expressed as follows:

$$\left(l_t, r_t\right) = \begin{cases} \left(l_t^{\varphi_j}, r_t^{\varphi_j}\right), & \text{if } \left(\text{score}_t^{\varphi_j} \geq \gamma_1\right) \text{ or} \\ & \left(\text{score}_t^{\varphi_j} < \gamma_1 \text{ and } \text{state}_{t-1} \right. \\ & \left. = \text{'not found'} \text{ and } E_t^j < \gamma_2\right) \\ \left(l_t^\phi, r_t^\phi\right), & \text{otherwise} \end{cases} \tag{18}$$

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Algorithm 2 MPC Procedure

Input: A satellite video sequence with T frames; The center coordinate (l_1, r_1) at first frame; $\gamma_1 = 0.4$; $\gamma_2 = 5$
Output: Predicted coordinates $\{(l_t, r_t)\}_{t=1}^{T}$

Initialize a Kalman filter through (l_1, r_1) and FC filter f through N template regions
Calculate $score_1^{\varphi_1}$ by (16); $state_1 \leftarrow$ "normal"
repeat
  Calculate $\{(l_t^{\varphi_i}, r_t^{\varphi_i})\}_{i=1}^{3}$ by (16); $(l_t^{\phi}, r_t^{\phi}) \leftarrow$ Kalman filter
  if $score_t^{\varphi_1} \geq \gamma_1$ **then**
    $j \leftarrow \arg\min_i E_t^i$ by (17)
    $(l_t, r_t) \leftarrow \text{Regression}\left[(l_t^{\varphi_j}, r_t^{\varphi_j})\right]$
    Update the Kalman filter using (l_t, r_t)
    if $j = 1$ **then**
      Update f with positive sample
      $state_t \leftarrow$ "normal"
    else
      Update f with hard negative sample
      $state_t \leftarrow$ "hard"
    end if
  else
    if $state_{t-1} =$ "normal" **then**
      $(l_t, r_t) \leftarrow (l_t^{\phi}, r_t^{\phi})$; $state_t \leftarrow$ "not found"
    else if $state_{t-1} =$ "not found" **then**
      $j \leftarrow \arg\min_i E_t^i$
      if $E_t^j < \gamma_2$ **then**
        $(l_t, r_t) \leftarrow \text{Regression}\left[(l_t^{\varphi_j}, r_t^{\varphi_j})\right]$
        Update the Kalman filter using (l_t, r_t)
        $state_{t-1} \leftarrow$ "normal"
        Update f with positive sample
      else
        $(l_t, r_t) \leftarrow (l_t^{\phi}, r_t^{\phi})$; $state_t \leftarrow$ "not found"
      end if
    end if
  end if
until end of sequence

here $j = \arg\min_i E_t^i$ defines the index of the predicted coordinate that is closest to the target motion trend. state_{t-1} is the target state of the frame t, so that when the target reappears after occluded, the model parameters can be adjusted in a timely manner. We set the thresholds for score γ_1 and correction value γ_2 to 0.4 and 5, respectively. Fig. 7 visualizes the target tracklet. It analyzes the target occlusion and reappearance situations. At this point, the advantage of calculating the correction value is demonstrated. It can more directly determine the tracked target's direction when it reappears and guide the tracker to successfully track the target. Algorithm 2 presents the overall flowchart of the tracker using the MPC scheme for online tracking. The target state, confidence score, and correction value are all involved in MPC module operation to achieve robust tracking. Fig. 8 shows the robustness of the proposed tracker with MPC module under occlusion attribute. The top figures show the tracking results in several frames, which visualize the search area and confidence map on the right side, respectively. The line chart below shows that the proposed tracker without an MPC module has failed to track since the first occlusion occurred.

IV. Experiments

The implementation details of our model are first introduced in this section. We then discuss the evaluation metrics on satellite video datasets. The proposed DTLTracker with ten state-of-the-art tracker models is evaluated in Section IV-C. The ablation studies are finally performed on three datasets.

A. Implementation Details

In network design, the input sizes of the template region and search area are both resized to 352 × 352. We use ResNet-50 [65] as the backbone network and set the FC filter size to 4 × 4. The target regression network uses the pretrained PreRoIPooling for bounding box regression which same as Baseline [27]. In the selection term of train datasets, first to 20th video sequences from VISO [11] and 120 sequences from SkySat [66] are used as training sets. Similar to Baseline, we utilize simultaneously LaSOT [67], GOT-10k [68], and TrackingNet [69] and synthesized images from COCO [70] dataset into the model training to increase the training samples. The validation sets use the other 17 sequences of SkySat, the 21st to 23rd video sequences of VISO, and partial GOT-10 k.

During offline training, the weights of the backbone network are fixed. DT, FC filter, and the regression branches are jointly trained [71]. Train-test pairs use a maximum interval of 500 frames within a video sequence. The iteration number of the proposed FC filter is set to 5. MPC module is related to target state and motion trajectory, so it does not participate during the training phase of target feature learning. The overall loss function is written as $\mathcal{L}_{\text{total}} = \omega_1\mathcal{L}(f) + \omega_2\mathcal{L}_{\text{res}}(\text{Regression}[(l_t^{\varphi_j}, r_t^{\varphi_j})])$. Our framework is trained 30 epochs on two GeForce RTX 2080 Ti with 5000 iterations for each epoch. The batch size is set to 10, and the learning rate to $1e{-}2$. In the online tracking phase, the initial FC filter iteration number is set to 30, with 10 and 1 for updating subsequent positive and hard negative samples, respectively. Data augmentations for the initial template samples include fliplr, rotate, blur, relativeshift, and dropout. The maximum number of decoded features stored during filter optimization is set to 50.

B. Evaluation Metrics

The one-pass evaluation (OPE) setting is adopted, that is, initialize the ground-truth position at the initial video frame. The precision and success plots, FPS, the area under the curve (AUC), and precision are set as the evaluation metrics in this article. The precision plots show the percentage of frames with a distance above a specified threshold to all video frames. The distance in each frame is the Euclidean distance between the ground truth and the tracked box center position. The curve is drawn by getting percentages with different thresholds. The success plots are expressed as the percentage of the intersection over union (IoU) ratio between

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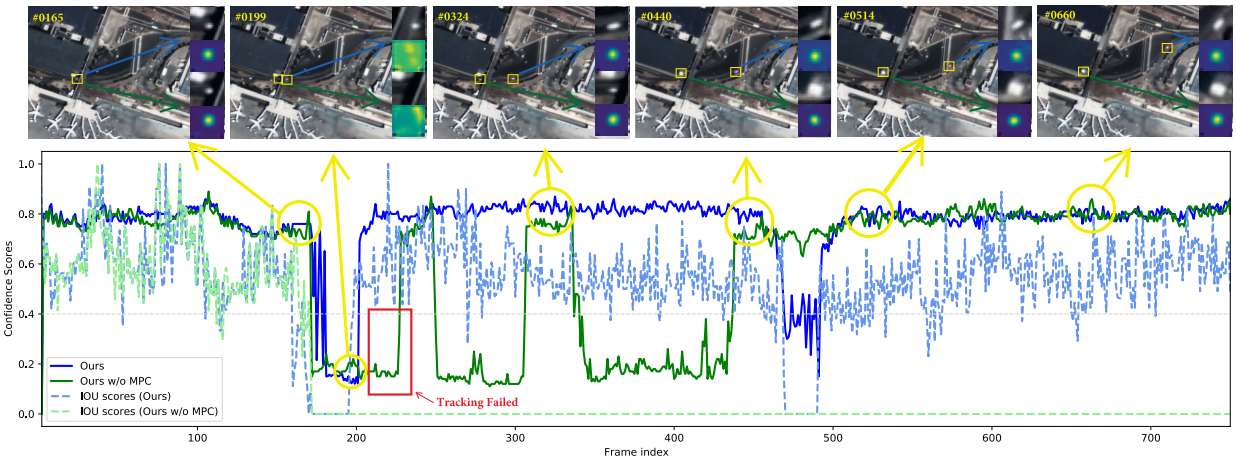


Fig. 8. Illustration of the IoU and predicted confidence scores at first of car sequence in SatSOT [12].

TABLE I

PERFORMANCE COMPARISON FOR TEN CHALLENGING ATTRIBUTES ON SV248S [7]. THE BEST PERFORMANCE IS HIGHLIGHTED IN **BOLD** FONT

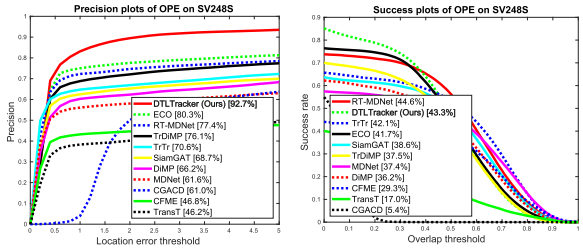
Model	STO		CO		LTO		SM		IV		IPR		DS		ND		BCL		BCH		
	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	FPS
CFME [56]	28.1	46.7	22.5	38.8	11.9	26.6	18.3	32.9	22.5	38.6	20.8	37.7	27.0	43.9	12.8	22.3	19.2	33.5	28.4	46.0	**228.7**
CGACD [72]	3.7	45.3	3.1	36.6	3.6	37.5	4.5	54.7	4.2	55.5	4.2	50.3	5.2	62.5	4.2	40.7	3.4	48.8	5.8	61.6	57.5
ECO [38]	31.2	63.3	24.0	48.7	24.5	58.3	38.7	77.8	36.5	73.2	36.9	74.9	41.5	81.2	39.3	75.7	**37.1**	75.4	40.9	78.0	28.9
MDNet [41]	24.5	45.5	19.0	31.5	35.2	63.8	**49.4**	82.8	39.2	64.9	32.5	58.9	38.0	63.7	**42.1**	70.2	28.5	51.6	33.6	54.4	3.8
RT-MDNet [42]	30.1	55.3	24.7	45.4	27.1	59.7	43.7	80.6	39.1	70.4	40.4	74.6	**45.4**	79.9	40.2	74.9	35.3	68.4	41.3	72.8	20.5
DiMP [27]	25.2	48.7	21.7	40.7	28.5	50.5	33.1	61.5	30.2	58.8	28.6	55.8	34.5	66.0	31.7	59.0	27.6	54.4	35.8	65.6	19.2
TrDiMP [29]	29.0	59.2	23.5	48.7	22.2	55.1	36.5	78.6	32.7	69.2	32.8	69.6	37.1	76.8	29.5	64.4	27.4	66.2	36.5	73.7	12.0
TransT [54]	12.8	37.9	8.9	30.7	14.7	47.5	12.1	35.7	11.4	36.5	13.1	41.7	14.1	43.8	4.1	16.5	8.1	32.6	20.3	50.4	23.0
SiamGAT [48]	29.6	54.1	25.3	45.4	32.8	58.2	34.3	64.0	35.0	65.0	35.2	63.7	37.4	68.8	33.9	62.9	32.3	61.5	37.3	65.3	37.0
TrTr [30]	26.8	49.0	21.7	40.5	34.4	57.7	40.7	72.9	**39.6**	68.1	36.3	64.1	41.9	72.1	38.9	67.7	32.1	60.2	40.9	65.7	31.8
DTLTraker (Ours)	**40.8**	**91.2**	**38.6**	**87.9**	**44.3**	**98.0**	43.9	**91.5**	37.7	**88.1**	**41.0**	**89.8**	41.7	**92.8**	35.4	**79.7**	35.9	**87.1**	**44.7**	**93.0**	11.4

the ground truth and the tracked bounding box is higher than a given threshold in all video sequences. The FPS evaluates the real-time performance of the trackers. It shows the response speed and operating efficiency. AUC, whose followed in the OTB evaluation metrics [73], [74], is defined as the AUC in the success plots. The precision represents the value when the threshold is 20 in the precision plots.

C. Evaluation on Satellite Video Datasets

We use three satellite datasets in this article to verify the effectiveness of the proposed tracker. It contains information-rich appearance descriptions with challenging scenes. These video sequences are labeled with accurate tracking targets, which are used for the evaluation of trackers. For comparison, we contrast the proposed DTLTracker with ten state-of-the-art tracking methods, include the correlation filter methods (CFME [56], ECO [38]), the deep convolutional networks (MDNet [41], RT-MDNet [42]), the Siamese networks (CGACD [72], DiMP [27], SiamGAT [48]), and the transformer trackers (TrDiMP [29], TransT [54], TrTr [30]). To get the best performance of these trackers, we test nine times for each satellite video sequence and take the middle results as the final for each tracker.

1) SV248S: This dataset utilizes 6 open-source satellite video datasets provided by CGSTL [7]. It constructs 248 video sequences. The frame rate is 25. Each dataset selects approximately 40 tracked objects including ships, motor vehicles, and aircraft. The resolution of these video sequences contains 3840 × 2160 and 4096 × 2160. Fig. 9 shows the overall success and precision plots. The baseline (DiMP) achieves a success score of 36.2% and a precision score of 66.2%. DTLTracker obtains the highest precision score 92.7% in Fig. 9(a). Compared with TrDiMP, TransT, and TrTr based on the Siamese network and transformer, our DTLTracker is more robust with the support of the proposed DT. In addition, the total frame number in the proposed DTLTracker with an IoU threshold less than 0.4 exceeds all state-of-the-art trackers shown in Fig. 9(b). The frames with IoU threshold less than 0.4 in DTLTracker are the most, about 70%.



(a) (b)

Fig. 9. (a) Precision and (b) success plots show a comparison of our DTLTracker with ten state-of-the-art trackers on SV248S [7].

Table I lists the AUC and precision scores of DTLTracker with ten state-of-the-art trackers for ten attributes in the SV248S dataset. These attributes include short-term

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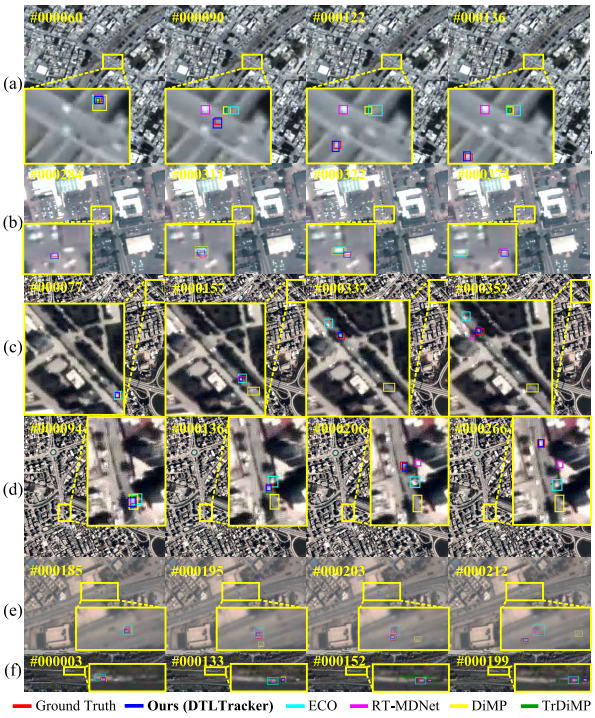


Fig. 10. Comparison of our proposed DTLTracker and the top-ranking trackers in six satellite video sequences. The video attributes from (a)–(f) represented in Table II, respectively.

TABLE II
LIST FOR SATELLITE VIDEO SEQUENCE ATTRIBUTES IN FIG. 10

Dataset	Video	Sequence	Attributes
SV248S	2	14	STO, LTO, DS, SM, CO, IPR
	5	18	STO, DS, IV, SM, IPR
VISO	26	180	-
	27	192	-
SatSOT	car	6	TO, LQ
	car	53	TO, SOB

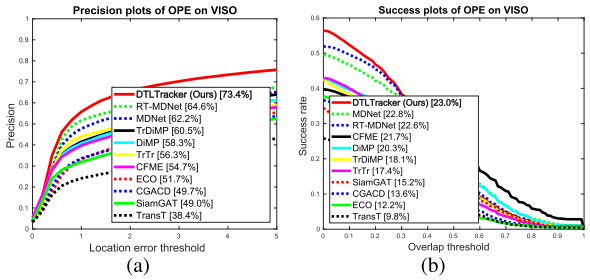


Fig. 11. (a) Precision and (b) success plots of the comparisons with state-of-the-art trackers on VISO [11].

occlusion (STO), continuous occlusion (CO), long-term occlusion (LTO), slow motion (SM), illumination variation (IV), in-plane rotation (IPR), dense similarity (DS), natural disturbance (ND), background cluster (BCL), and background change (BCH). DTLTracker achieves the best performance on STO, CO, LTO, IPR, and BCH attributes. Occlusion at STO, CO, and LTO attributes is a challenge that affects the tracked target feature learning. Our proposed MPC module has an impact on these occlusion attributes. It predicts the target trajectory better and optimizes the tracking effect when occlusion occurs. Besides, DTLTracker achieves a precision score of 8.7% higher than the second place MDNet [41] in the SM attribute. MDNet uses the single-branch network and performs well on its attribute. The IV attribute is of great significance to the verification of changes in learning target features. TrTr [30] and DTLTracker which adopt the transformer perform better, obtaining the highest AUC score of 39.6% and precision score of 88.1%, respectively. The DS attribute tests the performance of SOB interference. DTLTracker obtains a precision score of 92.8% and success of 41.7%. It suppresses the confidence scores of other objects across the proposed multiple cross-attention modules. The MPC module eliminates SOB interference through the correction value based on the Kalman filter in the post-processing stage.

Fig. 10 shows the visualization results of the proposed DTLTracker and the top-ranked trackers in some challenging satellite video sequences. Each original video frame has been visually enlarged to facilitate clearer observation of the tracking results. Table II lists the attributes of the corresponding satellite video sequences. Only DTLTracker tracks the target successfully after occluded by the bridge in Fig. 10(a). Other trackers stay at the original position after the occlusion occurs (like ECO [38], DiMP [27], TrDiMP [29]), or change the target to other objects (like RT-MDNet [42]). In Fig. 10(b), RT-MDNet distinguishes intraclass similarities effectively by the multitask loss. Our model tracks the target successfully through the DT and MPC models. The performance of DTLTracker is more robust than RT-MDNet at the video sequences with both occlusion and DS attributes.

2) VISO: This dataset is taken by Jilin-1, which contains some different traffic situations in real-world scenarios [11]. There are four common types of moving objects, including airplanes, cars, ships, and trains. A total of 27 video sequences is used for single object tracking tasks, which contain 3159 trajectories with 1.12 million frames. The resolution of these video sequences is uniformly set to 12 000 × 5000, and the frame rate is 10. For a fair comparison, we use the same video sequences as [11] for testing, namely videos 24–27 with 578 effective sequences. The precision and success plots of these 11 trackers are shown in Fig. 11. DTLTracker is 8.8% higher than the second place RT-MDNet [42] in the precision plot, obtaining a success score of 23.0%. The total frame number of DTLTracker whose IoU threshold is higher than 0.15 accounts for nearly half of the whole sequence in the success plot. DTLTracker demonstrates the strong robustness in Fig. 10(c). After short-term occlusion by trees, our DTLTracker tracks successfully with the assistance of the MPC model in Fig. 10(d). Other trackers (like ECO [38], TrDiMP [29]) track the target incorrectly as an object which looks similar to the target.

3) SatSOT: It uses the data collected by Jilin-1, Skybox, and Carbonite-2, which contains 105 video sequences with 27 664 frames [12]. The video frames for these sequences range from 120 to 750 frames. The tracked targets contain 26 ships, 65 cars, nine planes, and five trains, which sizes from 21 to 780 605 pixels. Fig. 12 demonstrates our DTLTracker outperforming to these state-of-the-art trackers. Table III lists the specific comparison results of accuracy and AUC scores between our proposed method and these

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TABLE III

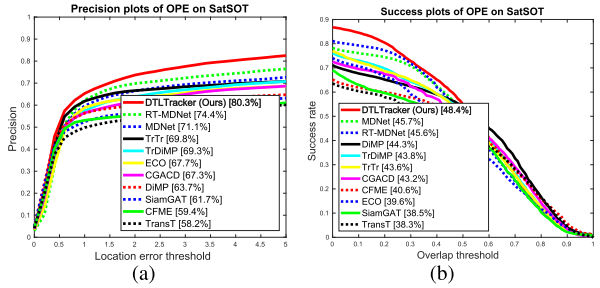
PERFORMANCE COMPARISONS OF OBJECT TRACKING METHODS FOR 11 ATTRIBUTES ON SATSOT [12]

Model	ARC		BC		BJT		POC		ROT		SOB		TO		DEF		FOC		IV		LQ		
	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	AUC (%)	Prec. (%)	FPS
CFME [56]	21.5	12.7	28.7	43.1	**59.9**	93.0	39.8	54.3	29.7	48.6	44.2	73.5	**39.3**	73.1	40.5	14.2	43.2	75.4	44.9	57.0	42.2	73.9	**206.0**
CGACD [72]	36.0	32.8	31.4	50.6	44.2	79.3	36.7	57.1	40.5	66.4	33.5	67.1	15.1	50.1	44.0	**30.8**	27.5	54.0	44.2	55.5	44.8	86.4	21.5
ECO [38]	30.5	23.0	27.6	49.0	40.0	84.5	31.9	51.3	33.5	64.0	32.7	76.4	18.3	65.9	41.1	21.1	29.3	63.3	43.5	70.8	46.9	97.5	20.6
MDNet [41]	43.4	38.0	37.8	58.2	47.4	81.8	35.6	50.8	37.9	64.0	37.3	72.0	32.1	71.5	41.7	21.5	34.7	66.0	50.2	71.1	**49.8**	95.8	3.8
RT-MDNet [42]	**45.9**	**42.8**	36.4	59.2	50.6	92.9	39.2	58.2	39.3	70.2	37.9	79.5	30.3	72.3	**46.3**	29.9	32.6	66.0	56.9	73.3	43.4	88.8	17.3
DiMP [27]	27.4	18.7	30.7	43.4	52.9	86.4	34.8	47.6	40.9	59.6	39.7	67.5	34.9	65.1	34.5	9.6	34.8	61.5	56.6	74.0	48.3	84.9	13.3
TrDiMP [29]	39.9	38.8	34.5	53.1	46.4	86.1	32.9	49.7	39.9	65.6	38.8	71.1	29.6	62.1	42.8	20.8	31.4	60.8	**59.1**	81.4	42.3	83.7	13.2
TransT [54]	31.6	25.7	22.4	33.5	37.5	65.5	31.6	45.4	33.7	54.9	27.0	57.8	14.4	48.1	37.3	19.6	36.8	69.7	46.7	54.9	26.4	54.7	9.5
SiamGAT [48]	18.2	17.7	26.6	42.7	38.7	68.3	29.6	48.7	39.1	61.6	36.5	65.2	31.9	68.5	22.0	8.5	31.4	60.5	45.7	67.2	44.1	74.7	46.6
TrTr [30]	29.5	31.0	32.2	55.1	45.7	83.8	31.4	51.2	41.9	67.9	38.8	72.6	27.6	70.2	29.0	28.5	32.5	61.1	39.4	54.7	47.8	97.9	13.8
DTLTraker (Ours)	41.8	40.6	**40.0**	**66.4**	45.0	**96.4**	**46.1**	**69.8**	**45.3**	**78.5**	**48.7**	**91.0**	38.8	**83.9**	39.8	23.8	**46.8**	**91.7**	55.3	**91.8**	48.6	**99.4**	10.6

TABLE IV

ABLATION STUDY ON SATELLITE DATASETS WITH SEVERAL CHALLENGING ATTRIBUTES

Model	DT	Filter	MPC	SV248S [7]									VISO [11]			SatSOT [12]								
				All			CO			DS						All			FOC			SOB		
				AUC (%)	Prec. (%)	FPS	AUC (%)	Prec. (%)	FPS	AUC (%)	Prec. (%)	FPS	AUC (%)	Prec. (%)	FPS	AUC (%)	Prec. (%)	FPS	AUC (%)	Prec. (%)	FPS	AUC (%)	Prec. (%)	FPS
Baseline	✘	Based	✘	34.8	66.0	18.2	17.0	43.4	20.1	19.3	65.3	18.0	18.7	55.6	20.3	43.2	63.8	16.4	33.0	61.0	18.7	32.3	64.6	17.6
	✘	**FC**	✘	36.6	66.7	16.3	23.0	46.2	17.9	28.6	66.5	16.3	18.7	51.3	21.1	43.9	64.0	16.6	33.7	61.1	19.0	33.2	70.0	17.9
$Ours_{MPC}$	✘	Based	✔	21.7	66.7	**24.4**	22.2	52.4	**24.0**	33.7	66.3	**24.6**	22.3	67.7	**22.8**	45.0	79.0	**19.8**	36.2	89.7	**20.7**	35.9	79.3	**20.6**
	✘	**FC**	✔	29.7	72.9	23.7	25.2	59.0	23.6	35.2	73.3	23.7	22.8	67.8	22.7	44.1	79.5	19.6	34.9	89.7	20.2	39.9	79.9	20.5
$Ours_{DT}$	✔	Based	✘	42.5	83.1	7.9	27.0	55.2	8.5	39.8	83.5	8.3	23.0	68.7	8.5	45.9	75.3	8.4	34.6	68.0	8.6	42.3	76.8	8.2
	✔	**FC**	✘	43.1	85.4	8.0	28.4	56.9	10.4	41.7	85.5	8.0	**23.5**	70.2	9.1	47.5	78.9	8.3	38.9	73.8	8.7	43.6	81.1	8.5
Ours	✔	Based	✔	41.3	92.1	12.8	36.1	87.4	14.2	41.5	92.1	12.8	22.3	71.7	18.5	44.0	77.6	11.0	44.8	89.6	9.8	43.2	81.1	10.3
	✔	**FC**	✔	**43.3**	**92.7**	11.4	**38.6**	**87.9**	12.7	**42.0**	**92.8**	11.5	23.0	**73.4**	16.3	**48.4**	**80.3**	10.6	**46.8**	**91.7**	11.4	**48.7**	**91.0**	10.5



(a) (b)

Fig. 12. (a) Precision and (b) success plots of the DTLTracker and ten trackers on SatSOT [12].

ten state-of-the-art trackers under the eleven attributes. These attributes are aspect ratio change (ARC), BC, background jitter (BJT), partial occlusion (POC), rotation (ROT), SOB, tiny object (TO), deformation (DEF), full occlusion (FOC), IV, and low quality (LQ). DTLTracker shows the state-of-the-art performance on SOB, FOC, BC, POC, and ROT attributes. It benefits from the attention learning of the target appearance by the DT module and the suppression of SOB interference by the MPC module. It can be seen that the MPC module is robust to the occlusion attributes on SatSOT. DTLTracker achieves tracking in low-pixel satellite scenarios compared to other trackers at Fig. 10(e). DiMP [27] and DTLTracker can discriminatively track the target at a similar background in Fig. 10(f), while other trackers (like ECO [38] and RT-MDNet [42]) are interfered by SOB.

D. Ablation Study

In this section, we conduct the ablation study from the individual modules of the proposed DT, filter optimization, and the MPC model. The proposed DTLTracker without our models (Baseline), with only DT ($Ours_{DT}$), with only MPC ($Ours_{MPC}$) and with our models (Ours) are compared under different filter optimization at three satellite video datasets. From Table IV, we implement the comparison between the AUC, precision, and FPS of these four trackers. It can be observed that our proposed tracker exhibits optimal performance in both overall and challenging attribute sequences. In addition, our trackers with MPC scheme accelerate by about 1–7 frames/s compared to those without this scheme. This is because the Kalman filter results in the MPC scheme will be adopted in some challenging sequences such as target occlusion. That is when the final coordinate is provided by the Kalman filter, the tracker can avoid the regression calculation. It achieves performance improvement on these challenging sequences while accelerating the online tracking process. The ablation experimental analysis is conducted from the following four aspects.

1) Discriminative Transformer: The evaluation results of $Ours_{DT}$ and Ours have surpassed the baseline and $Ours_{MPC}$ among all three satellite video datasets at Table IV. The DT serves as an important component for target temporal feature learning specific to remote sensing scenarios. It achieves feature enhancement and background suppression effectively. An experimental comparison of three proposed cross-attention modules is conducted in Table V, respectively. The same training datasets are adopted for all four trackers to achieve a fair comparison. It verifies that the fusion of multiple attention modules can effectively promote the target learning of the tracker on the premise of sacrificing running speeds.

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TABLE V
ABLATION STUDY ON DT WITHOUT MPC SCHEME UNDER FC FILTER

Model	DTA	DSA	DMA	SV248S AUC (%)	SV248S Prec. (%)	SV248S FPS	VISO AUC (%)	VISO Prec. (%)	VISO FPS	SatSOT AUC (%)	SatSOT Prec. (%)	SatSOT FPS
Ours_{DTA}	✔	✘	✘	38.4	76.0	10.2	18.5	59.4	11.6	46.9	69.5	9.7
Ours_{DSA}	✘	✔	✘	36.8	73.8	**10.4**	17.4	57.1	**12.0**	46.2	68.6	**10.0**
Ours_{DMA}	✘	✘	✔	40.0	78.8	9.6	19.2	61.2	10.2	47.1	70.0	9.2
Ours_{DT}	✔	✔	✔	**43.1**	**85.4**	8.0	**23.5**	**70.2**	9.1	**47.5**	**78.9**	8.3

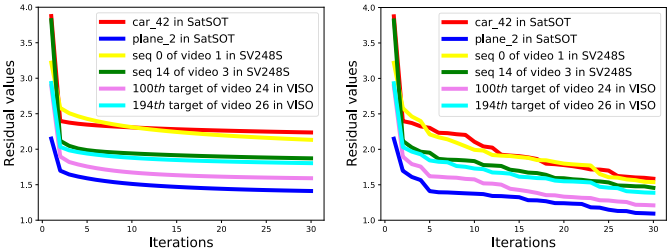


Fig. 13. Curve results of residual values with iterations in several satellite video sequences. (a) Based filter optimization. (b) FC filter optimization.

In addition, Ours_{DMA} uses product instead of addition in residual operation with increased time cost. Compared to the first two trackers, the running speed is about 1 frame/s lower.

2) Filter Optimization: The residual calculation function at ith iteration is set to be $\|\mathfrak{r}(\check{X} * f_i, \mathfrak{C})\|_2^2 + \|\lambda f_i\|_2^2$. Fig. 13 shows the residual values of different filter optimization under the various iteration values among six satellite video sequences. We set these two models to conduct experiments under DTL-Tracker at the online tracking stage. The step length α and optimization direction d_i are calculated through the neural network, which is efficient and fast. The proposed FC filter optimization requires no additional parameters. The convergence degree of the proposed optimization is significantly faster than the based filter. The evaluation results of FC-based trackers in Table IV on these three datasets are mostly better than those based filters with the same structures. It proves that the tracker can locate the target accurately across the premise of the FC filter among the same iterations.

3) MPC Scheme: This mechanism is a post-processing process that uses the Kalman filter to predict target motion. It avoids tracking failures effectively under occlusion or SOB influence, as well as accelerates the online process. In Table IV, Ours_{MPC} and Ours with the MPC module exhibit significantly higher performance in occlusion and similarity interference attributes compared to Baseline and Ours_{DT}. It indicates that this module can effectively learn the temporal motion trend of the target. In addition, we conduct an ablation experiment for γ_1 and γ_2 on the occlusion and DS attributes of SV248S with the proposed Transformer at Table VI. The smaller the values of γ_1 and γ_2 are, the better tracker is not necessarily. The tracker with γ_1 of 0.4 performs better than others. And its performance is optimal when γ_2 is 5. Under the premise with state_{t-1} = "normal," as γ_1 increases, the premise that state_t = "not found" decreases. At this point, the earlier the Kalman filter results (l_t^{ϕ}, r_t^{ϕ}) is activated as the final target center, the faster the tracker runs. Due to the direct correlation between γ_2 and Euclidean distance E_t^j, the larger the threshold set by γ_2, the earlier the model will adopt the confidence result $(l_t^{\varphi_j}, r_t^{\varphi_j})$ as the final target center when state_{t-1} = "not found." It means executing the Kalman filter will end earlier, which slows down the running speed. Therefore, the results show that the speed gradually decreases with the decrease of γ_1 and the increase of γ_2 in Table VI. Besides, MPC shows strong post-processing capabilities at the target reappearing after occlusion of Fig. 14(d).

TABLE VI
ABLATION STUDY ON MPC SCHEME WITH THE PROPOSED TRANSFORMER AND FC FILTER

γ_1	γ_2	CO AUC (%)	CO Prec. (%)	CO FPS	LTO AUC (%)	LTO Prec. (%)	LTO FPS	STO AUC (%)	STO Prec. (%)	STO FPS	DS AUC (%)	DS Prec. (%)	DS FPS
0.4	2	7.9	33.7	**62.8**	24.8	67.3	**36.3**	8.9	37.5	**57.7**	11.0	44.1	**54.7**
0.4	3	19.3	60.7	32.6	29.5	71.0	21.4	18.2	61.7	32.4	21.2	66.5	31.1
0.2	4	27.3	75.6	14.0	24.6	77.1	12.5	27.9	78.7	13.0	30.6	84.0	13.7
0.3	4	24.2	75.0	18.7	22.0	80.5	16.0	25.3	78.6	17.9	26.8	84.2	18.1
0.4	4	23.7	69.4	20.7	31.7	82.1	17.0	23.7	71.3	20.5	27.2	76.8	21.0
0.5	4	21.4	70.6	27.5	30.1	83.4	19.9	19.4	68.6	29.4	21.7	73.0	29.5
0.2	5	31.1	81.2	9.7	31.5	81.7	7.5	31.8	83.1	9.2	34.7	86.4	9.3
0.3	5	31.3	78.5	11.7	38.0	90.5	9.5	32.5	81.6	11.4	35.2	85.0	11.0
0.4	5	**38.6**	**87.9**	12.7	44.3	98.0	10.9	**40.8**	**91.2**	12.2	**41.7**	**92.8**	11.5
0.5	5	24.8	73.2	18.4	37.8	93.4	11.0	24.9	74.0	21.9	28.2	78.0	21.1
0.2	6	33.8	82.8	7.8	34.3	85.8	7.1	36.3	85.6	6.6	38.7	89.9	7.5
0.3	6	34.1	80.1	8.3	38.0	88.3	7.5	35.0	84.1	7.5	37.9	88.2	8.6
0.4	6	34.2	81.7	10.2	43.5	99.7	8.1	33.6	82.6	11.3	35.4	84.1	12.2
0.5	6	31.2	75.8	18.8	40.7	92.4	8.8	29.8	74.3	20.0	32.8	78.8	19.1
0.2	7	35.3	81.2	6.2	33.8	78.1	5.5	38.6	87.1	5.2	40.3	89.2	6.4
0.3	7	35.7	82.3	7.8	39.3	86.5	6.6	38.8	88.0	6.2	40.0	90.7	7.1
0.4	7	35.9	83.0	8.6	**45.2**	97.1	7.0	35.2	80.8	9.6	37.9	86.4	10.1
0.5	7	33.7	76.1	13.2	42.8	92.4	8.2	33.2	77.6	14.9	35.0	80.5	16.1
0.4	8	33.9	78.9	17.5	42.1	96.3	7.5	34.0	78.7	9.0	36.5	83.0	8.1
0.4	9	34.1	76.1	5.5	38.8	85.1	6.0	35.0	77.6	7.1	36.3	81.6	9.0
0.4	10	33.2	71.5	6.4	36.0	77.2	6.0	34.1	74.2	7.7	36.3	79.7	9.0

TABLE VII
LIST FOR SATELLITE VIDEO SEQUENCE ATTRIBUTES IN FIG. 14

Dataset	Video	Sequence	frame	Attributes
SV248S	1	3	112th	DS, IPR
	1	8	64th	DS, IV, BCH
	1	44	249th	DS, IV, BCH, LTO, IPR, BCL
SatSOT	car	1	214th	ROT, FOC
	car	17	87th	ROT, SOB, BC

4) Visual Demonstration: The confidence maps of these trackers on some satellite videos have been shown in Fig. 14. The corresponding sequence attributes are shown in Table VII. Our tracker obtains a response map discriminatively with almost no influence from other SOBs and BC. Fig. 14(d) shows the scores of the target reappearing after occlusion. Ours_{MPC} and Our tracker achieve tracking successfully with max scores higher than 0.8. From the overall view of Fig. 14, Ours and Ours_{DT} almost outperform the baseline and Ours_{MPC} significantly in target feature learning. It demonstrates the developed discriminative temporal learning Transformer has great advantages in utilizing target spatiotemporal information.

E. Limitations Discussion

Effective spatiotemporal feature learning relies on the DT. But its attention calculation increases memory burden as well as hinders the real-time performance of the running speed.

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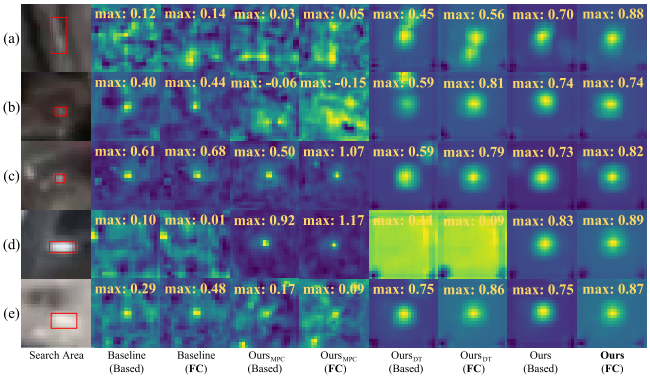


Fig. 14. Visualization of the confidence maps for the baseline, Ours$_{MPC}$, Ours$_{DT}$, and Our proposed tracker across different optimizations on five satellite videos. The video attributes from (a)–(e) represented in Table VII, respectively.

In future work, we can explore the sparse attention to reduce memory burden. The regression of the tracked target box is not very accurate and thus reduces the AUC constructed from the IoU ratio. Therefore, future research will introduce multiscale representation learning into the target regression module to make the model more robust.

V. Conclusion

This article constructs a DTL tracker, which shows high efficiency and robustness in learning the target spatiotemporal features among satellite videos. The proposed discriminative fusion decoder provides an effective temporal feature learning framework for transformer tracking. It supplies a solution for SOB interference in satellite scenes. In addition, an FC filter optimization with the conjugate gradient method provides a prerequisite for rapid convergence. It promotes the model to achieve optimal iteration in the offline training and online tracking phase. The MPC mechanism utilizes the temporal information of the sequence to capture motion trends, assuring successful tracking in complex satellite scenes during the post-processing stage. In future research, considering the diversity of feature learning, we will focus on achieving the real-time multiscale features learning at satellite video tasks.

References

[1] X. Yang et al., "Relation learning reasoning meets tiny object tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, 2024.

[2] W. Zhang, W. Deng, Z. Cui, J. Liu, and L. Jiao, "Object knowledge distillation for joint detection and tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024.

[3] H. Guo, X. Su, C. Wu, B. Du, and L. Zhang, "Building-road collaborative extraction from remote sensing images via cross-task and cross-scale interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, 2024.

[4] W. Zhang, L. Jiao, F. Liu, L. Li, X. Liu, and J. Liu, "MBLT: Learning motion and background for vehicle tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[5] X. Li et al., "A complex-former tracker with dynamic polar spatio-temporal encoding," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, pp. 1–15, 2023.

[6] J. Feng, Y. Liang, X. Zhang, J. Zhang, and L. Jiao, "SDANet: Semantic-embedded density adaptive network for moving vehicle detection in satellite videos," *IEEE Trans. Image Process.*, vol. 32, pp. 1788–1801, 2023.

[7] Y. Li et al., "Deep learning-based object tracking in satellite videos: A comprehensive survey with a new dataset," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 4, pp. 181–212, Dec. 2022.

[8] W. Ao, Y. Fu, X. Hou, and F. Xu, "Needles in a haystack: Tracking city-scale moving vehicles from continuously moving satellite," *IEEE Trans. Image Process.*, vol. 29, pp. 1944–1957, 2020.

[9] X. Chen, H. Sui, J. Fang, M. Zhou, and C. Wu, "A novel AMS-DAT algorithm for moving vehicle detection in a satellite video," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[10] I. Delibasoglu, "PESMOD: Small moving object detection benchmark dataset for moving cameras," in *Proc. 7th Int. Conf. Frontiers Signal Process. (ICFSP)*, Sep. 2022, pp. 23–29.

[11] Q. Yin et al., "Detecting and tracking small and dense moving objects in satellite videos: A benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[12] M. Zhao, S. Li, S. Xuan, L. Kou, S. Gong, and Z. Zhou, "SatSOT: A benchmark dataset for satellite video single object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[13] Z. Pi et al., "Very low-resolution moving vehicle detection in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[14] P. He et al., "MANet: Multi-scale aware-relation network for semantic segmentation in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[15] Y. Zhou, S. Chen, J. Zhao, R. Yao, Y. Xue, and A. E. Saddik, "CLT-Det: Correlation learning based on transformer for detecting dense objects in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[16] Y. Zhou, F. Wang, J. Zhao, R. Yao, S. Chen, and H. Ma, "Spatial–temporal based multihead self-attention for remote sensing image change detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6615–6626, Oct. 2022.

[17] L. Jiao et al., "Transformer meets remote sensing video detection and tracking: A comprehensive survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1–45, 2023.

[18] W. Song et al., "A joint Siamese attention-aware network for vehicle object tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[19] X. Yang, Y. Wang, N. Wang, and X. Gao, "An enhanced SiamMask network for coastal ship tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[20] J. Shao, B. Du, C. Wu, M. Gong, and T. Liu, "HRSiam: High-resolution Siamese network, towards space-borne satellite video tracking," *IEEE Trans. Image Process.*, vol. 30, pp. 3056–3068, 2021.

[21] L. Jiao, D. Wang, Y. Bai, P. Chen, and F. Liu, "Deep learning in visual tracking: A review," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 9, pp. 5497–5516, Sep. 2021.

[22] Y. Nie, C. Bian, and L. Li, "Object tracking in satellite videos based on Siamese network with multidimensional information-aware and temporal motion compensation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[23] L. Ruan, Y. Guo, D. Yang, and Z. Chen, "Deep Siamese network with motion fitting for object tracking in satellite videos," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[24] Y. Wang, S. Mei, S. Zhang, and Q. Du, "SiamMRAAN: Siamese multi-level residual attention adaptive network for hyperspectral videos tracking," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 5275–5278.

[25] C. Fu, W. Peng, S. Li, J. Ye, and Z. Cao, "Local perception-aware transformer for aerial tracking," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2022, pp. 12122–12129.

[26] M. Danelljan, G. Bhat, F. S. Khan, and M. Felsberg, "ATOM: Accurate tracking by overlap maximization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4660–4669.

[27] G. Bhat, M. Danelljan, L. Van Gool, and R. Timofte, "Learning discriminative model prediction for tracking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6181–6190.

[28] M. Danelljan, L. Van Gool, and R. Timofte, "Probabilistic regression for visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 7183–7192.

[29] N. Wang, W. Zhou, J. Wang, and H. Li, "Transformer meets tracker: Exploiting temporal context for robust visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1571–1580.

[30] M. Zhao, K. Okada, and M. Inaba, "TrTr: Visual tracking with transformer," 2021, *arXiv:2105.03817*.

[31] S. Li, F. Liu, and L. Jiao, "Self-training multi-sequence learning with transformer for weakly supervised video anomaly detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 2, 2022, pp. 1395–1403.

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[32] J. Selva, A. S. Johansen, S. Escalera, K. Nasrollahi, T. B. Moeslund, and A. Clapés, "Video Transformers: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 12922–12943, Nov. 2023.

[33] P. Lai, M. Zhang, G. Cheng, S. Li, X. Huang, and J. Han, "Target-aware transformer for satellite video object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–10, 2024.

[34] J. R. Shewchuk et al., "An introduction to the conjugate gradient method without the agonizing pain," School Comput. Sci., Carnegie Mellon Univ., Pittsburgh, PA, USA, Tech. Rep., 1994.

[35] D. S. Bolme, J. R. Beveridge, B. A. Draper, and Y. M. Lui, "Visual object tracking using adaptive correlation filters," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 2544–2550.

[36] J. F. Henriques, R. Caseiro, P. Martins, and J. Batista, "High-speed tracking with kernelized correlation filters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 583–596, Mar. 2015.

[37] W. Zhang, L. Jiao, F. Liu, S. Yang, and J. Liu, "DFAT: Dynamic feature-adaptive tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 1, pp. 43–58, Jan. 2023.

[38] M. Danelljan, G. Bhat, F. S. Khan, and M. Felsberg, "ECO: Efficient convolution operators for tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6931–6939.

[39] X. Zhang et al., "A deep learning filter for visual drone single object tracking," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 638–650.

[40] M. Danelljan, A. Robinson, F. Shahbaz Khan, and M. Felsberg, "Beyond correlation filters: Learning continuous convolution operators for visual tracking," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 472–488.

[41] H. Nam and B. Han, "Learning multi-domain convolutional neural networks for visual tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4293–4302.

[42] I. Jung, J. Son, M. Baek, and B. Han, "Real-time MDNet," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 83–98.

[43] L. Bertinetto, J. Valmadre, J. F. Henriques, A. Vedaldi, and P. H. S. Torr, "Fully-convolutional Siamese networks for object tracking," in *Proc. Eur. Conf. Comput. Vis.* Springer, 2016, pp. 850–865.

[44] B. Li, J. Yan, W. Wu, Z. Zhu, and X. Hu, "High performance visual tracking with Siamese region proposal network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8971–8980.

[45] G. Chen, P. Zhu, B. Cao, X. Wang, and Q. Hu, "Cross-drone transformer network for robust single object tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 9, pp. 4552–4563, Jun. 2023.

[46] J. Valmadre, L. Bertinetto, J. Henriques, A. Vedaldi, and P. H. S. Torr, "End-to-end representation learning for correlation filter based tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 5000–5008.

[47] J. Nie, H. Wu, Z. He, M. Gao, and Z. Dong, "Spreading fine-grained prior knowledge for accurate tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 9, pp. 6186–6199, Sep. 2022.

[48] D. Guo, Y. Shao, Y. Cui, Z. Wang, L. Zhang, and C. Shen, "Graph attention tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9543–9552.

[49] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lucic, and C. Schmid, "ViViT: A video vision transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6836–6846.

[50] S. W. Oh, J.-Y. Lee, N. Xu, and S. J. Kim, "Video object segmentation using space-time memory networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9226–9235.

[51] S. Yun, J. Kim, D. Han, H. Song, J.-W. Ha, and J. Shin, "Time is MattEr: Temporal self-supervision for video transformers," in *Proc. 39th Int. Conf. Mach. Learn.*, Jul. 2022, pp. 25804–25816.

[52] H. Guo, X. Su, C. Wu, B. Du, and L. Zhang, "SAAN: Similarity-aware attention flow network for change detection with VHR remote sensing images," *IEEE Trans. Image Process.*, vol. 33, pp. 2599–2613, 2024.

[53] Z. Yang, Y. Wei, and Y. Yang, "Associating objects with transformers for video object segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 2491–2502.

[54] X. Chen, B. Yan, J. Zhu, D. Wang, X. Yang, and H. Lu, "Transformer tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8126–8135.

[55] J. Shao, B. Du, C. Wu, and L. Zhang, "Can we track targets from space? A hybrid kernel correlation filter tracker for satellite video," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8719–8731, Nov. 2019.

[56] S. Xuan, S. Li, M. Han, X. Wan, and G.-S. Xia, "Object tracking in satellite videos by improved correlation filters with motion estimations," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1074–1086, Feb. 2020.

[57] Y. Guo, D. Yang, and Z. Chen, "Object tracking on satellite videos: A correlation filter-based tracking method with trajectory correction by Kalman filter," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3538–3551, Sep. 2019.

[58] J. Shao, B. Du, C. Wu, and Y. Pingkun, "PASiam: Predicting attention inspired Siamese network, for space-borne satellite video tracking," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2019, pp. 1504–1509.

[59] Y. Cui, C. Jiang, L. Wang, and G. Wu, "MixFormer: End-to-end tracking with iterative mixed attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13608–13618.

[60] S. Gao, C. Zhou, C. Ma, X. Wang, and J. Yuan, "AiATrack: Attention in attention for transformer visual tracking," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 146–164.

[61] H. Zhang, Y. Hao, and C.-W. Ngo, "Token shift transformer for video classification," in *Proc. 29th ACM Int. Conf. Multimedia*, Oct. 2021, pp. 917–925.

[62] A. Bulat, J. M. P. Rua, S. Sudhakaran, B. Martinez, and G. Tzimiropoulos, "Space-time mixing attention for video transformer," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 19594–19607.

[63] C. Liu, H. Yang, J. Fu, and X. Qian, "Learning trajectory-aware transformer for video super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5687–5696.

[64] J. Nocedal and S. J. Wright, *Numerical Optimization*. New York, NY, USA: Springer, 1999.

[65] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[66] *Planet Application Program Interface: In Space for Life on Earth*, Planet Team, San Francisco, CA, USA, 2017.

[67] H. Fan et al., "LaSOT: A high-quality benchmark for large-scale single object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5369–5378.

[68] L. Huang, X. Zhao, and K. Huang, "GOT-10k: A large high-diversity benchmark for generic object tracking in the wild," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 5, pp. 1562–1577, May 2019.

[69] M. Müller, A. Bibi, S. Giancola, S. Alsubaihi, and B. Ghanem, "TrackingNet: A large-scale dataset and benchmark for object tracking in the wild," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 310–327.

[70] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, Zurich, Switzerland. Springer, 2014, pp. 740–755.

[71] B. Jiang, R. Luo, J. Mao, T. Xiao, and Y. Jiang, "Acquisition of localization confidence for accurate object detection," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2018, pp. 784–799.

[72] F. Du, P. Liu, W. Zhao, and X. Tang, "Correlation-guided attention for corner detection based visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6836–6845.

[73] Y. Wu, J. Lim, and M.-H. Yang, "Online object tracking: A benchmark," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 2411–2418.

[74] Y. Wu, J. Lim, and M.-H. Yang, "Object tracking benchmark," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1834–1848, Sep. 2015.



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